# Erdős Problem #1063 for prime $k$: an exact $q$-adic criterion and the lower bound $n_p \geq p^{11/6-\varepsilon}$

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**Problem.** For $k \geq 2$ let $n_k \geq 2k$ be the least $n$ such that $n-i \mid \binom{n}{k}$ for all but one $0 \leq i < k$. Estimate $n_k$.
**Known.** $n_k \leq k!$ (Monier) and $n_k \leq k \cdot \operatorname{lcm}(2,\dots,k-1) = e^{(1+o(1))k}$ (Cambie). For prime $k=p$ the available lower bounds give only $n_p \geq p$ up to constants.
**Results here.** For prime $k=p$: an exact criterion for admissibility (Theorem 3); the unconditional bound $n_p \geq p^{11/6-\varepsilon}$ (and $p^{39/20-\varepsilon}$ using Jia's theorem), and $n_p \geq p^{2-o(1)}$ under RH (Theorem 8); and a heuristic, supported by the data, that $\log n_p \asymp p/\log p$. In particular $n_p$ is very probably *not* bounded by a polynomial in $p$.

Throughout, $p \geq 5$ is prime, $n \geq 2p$, and we write

$$n = Mp + i_0, \qquad 0 \leq i_0 < p, \qquad M = \lfloor n/p \rfloor \geq 2.$$

Call $n$ *admissible* if $n-i \mid \binom{n}{p}$ for all but one $0 \leq i < p$. The block is $\mathcal{B} = \{n-i : 0 \leq i < p\}$.

## 1 Reduction to valuations

Clearly $n-i \mid \binom{n}{p}$ if and only if $v_q(n-i) \leq v_q\binom{n}{p}$ for every prime $q$. For $q > p$ we have $q \nmid p!$, so $v_q\binom{n}{p} = \sum_j v_q(n-j) \geq v_q(n-i)$. Hence only primes $q \leq p$ matter. We use Kummer's theorem in the form

$$v_q\binom{n}{p} = \#\{\, j \geq 1 :\ n \bmod q^j < p \bmod q^j \,\}. \tag{1}$$

**Lemma 1** (the excluded index)**.** $v_p\binom{n}{p} = v_p(n-i_0) - 1$. *Consequently* $n - i_0 \nmid \binom{n}{p}$*, and the excluded index of an admissible* $n$ *is always* $i_0 = n \bmod p$.

*Proof.* In (1) with $q=p$, the case $j=1$ never counts since $p \bmod p = 0$. For $j \geq 2$ we have $p \bmod p^j = p$. Also $n \bmod p^j < p$ holds if and only if $n \bmod p^j = i_0$, i.e. $p^j \mid n - i_0$. So the count is $v_p(n-i_0) - 1$. Every other $n-i$ is prime to $p$. □

So $n$ is admissible if and only if, for every prime $q < p$ and every $i \neq i_0$, $v_q(n-i) \leq v_q\binom{n}{p}$.

## 2 The exact criterion

For a prime $q < p$ put $L_q = \lfloor \log_q p \rfloor \geq 1$. Define

$$e_q = \max_{0 \leq i < p} v_q(n-i), \qquad e_q^* = \max_{i \neq i_0} v_q(n-i), \qquad D_q = \#\{\, 1 \leq j \leq L_q :\ n \bmod q^j \geq p \bmod q^j \,\}.$$

**Lemma 2.** $v_q\binom{n}{p} = e_q - D_q$.

*Proof.* $\mathcal{B}$ consists of $p \geq q^{L_q}$ consecutive integers, so it contains a multiple of $q^{L_q}$, and $e_q \geq L_q$. For $j > L_q$ we have $p \bmod q^j = p$. Then $n \bmod q^j < p$ if and only if $\mathcal{B}$ contains a multiple of $q^j$, i.e. $j \leq e_q$. So the indices $j > L_q$ contribute $e_q - L_q$ to (1). The indices $j \leq L_q$ contribute $L_q - D_q$ by definition. □

**Theorem 3** (criterion)**.** *$n \geq 2p$ is admissible if and only if, for every prime $q < p$,*

$$D_q = 0 \qquad or \qquad v_q(n - i_0) \;\geq\; e_q^* + D_q.$$

*Proof.* By Lemmas 1 and 2, admissibility means $e_q^* \leq e_q - D_q$ for all $q < p$. If $D_q = 0$ this always holds. If $D_q \geq 1$ it forces $e_q^* < e_q$. Then the maximum $e_q$ is attained only at $i_0$, and the condition reads $v_q(n - i_0) \geq e_q^* + D_q$. □

**Corollary 4** (large primes)**.** *Let $q$ be a prime with $p/2 < q < p$. Then $q$ satisfies the condition of Theorem 3 if and only if*

$$(a)\;\; n \bmod q < p - q \qquad or \qquad (b)\;\; q \mid M.$$

*Proof.* Here $q^2 > p$, so $L_q = 1$ and $p \bmod q = p - q$. Thus $D_q = 0$ is exactly (a). If (a) fails, write $r = n \bmod q \geq p - q$. Then $r + q \geq p$, so $n - r$ is the only multiple of $q$ in $\mathcal{B}$. The condition $v_q(n - i_0) \geq e_q^* + 1$ then holds if and only if this multiple is $n - i_0 = pM$. Since $q \neq p$, that means $q \mid M$. □

**Verification.** We checked Theorem 3 against the definition, computing $\binom{n}{p}$ valuations directly. The check covered every $n$ with $2p \leq n < 20000$ and every $p \in \{5, 7, 11, 13, 17, 19, 23\}$: 139,810 pairs, of which 7,931 are admissible, with no mismatch.

# 3 Lower bounds up to the $p^2$ barrier

## 3.1 The primes $q \in (p/2, p)$

**Lemma 5** (structure)**.** *Assume $M < p/2$. Let $q = p - g$ be a prime with $1 \leq g < p/2$. Then $q$ satisfies the condition of Theorem 3 if and only if*

$$g \in \bigcup_{j \geq 1} I_j, \qquad I_j = [a_j, b_j), \quad a_j = \frac{jp - i_0}{M + j}, \quad b_j = \frac{jp - i_0}{M + j - 1}.$$

*Proof.* We have $q > p/2 > M \geq 2$, so $q \nmid M$. By Corollary 4 the condition is (a): $n \bmod q < g$. Write $n = Mq + s$ with $s = Mg + i_0$, and put $j = \lfloor s/q \rfloor$. Then

$$n \bmod q = s - jq = (M + j)g + i_0 - jp.$$

So (a) holds if and only if $0 \leq (M + j)g + i_0 - jp < g$, i.e. $g \in I_j$. Conversely, if $g \in I_j$ for some $j$, then $s - jq \in [0, g) \subset [0, q)$, so this $j$ equals $\lfloor s/q \rfloor$ and (a) holds. Finally $j = 0$ is impossible, since it would need $(M - 1)g < -i_0$. □

**Verification.** We checked Lemma 5 for every prime $11 \leq p < 200$, every $2p \leq n < 40p$ and every prime $q \in (p/2, p)$: 2,091,482 triples, with no disagreement.

**Lemma 6** (many prime-free gaps)**.** *Let $n$ be admissible with $8 \le M < p/2$. For $1 \le j \le \lfloor M/4 \rfloor$ the intervals $J_j = [b_j, a_{j+1})$ are pairwise disjoint and contained in $(0, p/2)$. Each has length at least $\frac{2p}{9M}$, and no $g \in J_j$ makes $p - g$ prime.*

*Proof.* A direct computation gives

$$a_{j+1} - b_j = \frac{p(M-1-j) + 2i_0}{(M+j+1)(M+j-1)} \;\ge\; \frac{p(M-1-j)}{(M+j+1)^2}.$$

For $j \le M/4$ and $M \ge 8$ we have $M-1-j \ge M/2$ and $M+j+1 \le 3M/2$, so the length is at least $\frac{2p}{9M}$. Also $a_{j+1} < \frac{(j+1)p}{M+j+1} < \frac{p}{2}$ because $j+1 < M$. The intervals are ordered as $a_1 < b_1 < a_2 < b_2 < \cdots$, so the $J_j$ are disjoint gaps between consecutive $I_j$. By Lemma 5, a prime $p - g$ with $g \in J_j$ would violate admissibility. □

**Lemma 7** (small $M$)**.** *For all sufficiently large $p$, $M = \lfloor n_p/p \rfloor \ge 8$.*

*Proof.* Suppose $2 \le M < 8$. The set $[1, a_1) \cup [b_1, p/(2M))$ consists of gaps of Lemma 5. Note $p/(2M) \le a_2$, since $M \ge 2$, and both pieces lie in $g < p/4$, i.e. $q \in (3p/4, p)$. Put $A = a_1$, $B = b_1$, $C = p/(2M)$. If $B \le C$, the total length is $(A-1) + (C-B) = C - (B-A) - 1$. Since $B - A = \frac{p - i_0}{M(M+1)} \le \frac{p}{M(M+1)}$, this is at least $\frac{p(M-1)}{2M(M+1)} - 1 \ge \frac{p}{6M} - 1$. If $B > C$, then $p - i_0 > p/2$, so $\min(A, C) - 1 \ge \frac{p}{3M} - 1$. Either way one piece has length at least $\frac{p}{12M} - 1 > \frac{p}{100}$. So $(3p/4, p)$ would contain a prime-free interval of length $p/100$, which contradicts the prime number theorem for large $p$. □

### 3.2 The bound

For $0 < \theta < 1$ say that $\theta$ is *admissible for short intervals* if

$$\#\{\, x \le X : \; (x,\, x + x^\theta] \text{ contains no prime} \,\} = o(X) \qquad (X \to \infty).$$

Huxley (1972) showed every $\theta > 1/6$ is admissible.

**Theorem 8.** *If $\theta$ is admissible for short intervals, then $n_p \ge p^{2-\theta}/10$ for all sufficiently large primes $p$. In particular, unconditionally,*

$$n_p \;\ge\; p^{11/6 - \varepsilon} \qquad (p \ge p_0(\varepsilon)).$$

*Proof.* Let $n = n_p$. If $M \ge p/2$ then $n \ge p^2/2$. Otherwise $8 \le M < p/2$ by Lemma 7. Let $\ell = \frac{2p}{9M}$ and suppose $\ell \ge 2p^\theta + 6$.

Each gap $J_j$ from Lemma 6 corresponds to a prime-free interval of $q = p - g$ inside $(p/2, p)$ of length at least $\ell$. It contains at least $\ell - p^\theta - 3 \ge \ell/2$ integers $x$ with $(x, x + p^\theta]$ inside the same interval, so $(x, x + x^\theta]$ has no prime. There are at least $\lfloor M/4 \rfloor \ge M/8$ disjoint gaps. This gives at least $\frac{M}{8} \cdot \frac{p}{9M} = \frac{p}{72}$ exceptional integers $x \in (p/2, p)$, contradicting admissibility of $\theta$ with $X = p$.

Hence $\ell < 2p^\theta + 6$, i.e. $M > \frac{p}{9(p^\theta + 3)} \ge \frac{p^{1-\theta}}{10}$ for large $p$, and $n \ge pM \ge p^{2-\theta}/10$. □

*Remark.* (1) The single gap below $a_1$ together with Baker–Harman–Pintz already gives $n_p \gg p^{1.475}$. Theorem 8 uses all $\asymp M$ gaps at once.

(2) Any almost-all exponent transfers directly. Jia (1996) proved that every $\theta > 1/20$ is admissible, which gives $n_p \ge p^{39/20 - \varepsilon}$ unconditionally. A recent preprint of R. Li (arXiv:2407.05651) claims $\theta > 1/21.5$, which would give $n_p \ge p^{2 - 2/43 - \varepsilon}$. Huxley's exponent is used in the theorem

because it is the classical, fully standard input. Under RH, Selberg's theorem (almost all intervals $(x, x + f(x)\log^2 x]$ contain a prime, for any $f \to \infty$) gives $n_p \gg p^2/(f(p)\log^2 p)$.

(3) The method stops at $p^2$. Once $M \geq p$, the gaps $J_j$ have length below 1. Beyond that point, sieve-type upper bounds (e.g. the large sieve) only show that admissible $n \leq X$ are rare, not absent, so a super-polynomial lower bound needs a new idea.

## 4 Heuristic size and data

By Theorem 3, admissibility is essentially the event "$D_q = 0$ for all $q < p$", except for a few primes paid for by divisibility of $M$. For $\sqrt{p} < q < p$ this event is $n \bmod q < p \bmod q$. For $q \in (p/(m+1), p/m]$ the ratio $(p \bmod q)/q$ is roughly uniform on $(0, 1)$, so each such prime costs a factor about $e^{-1}$ on average. The primes $q \leq \sqrt{p}$ contribute $e^{O(\sqrt{p})}$. Treating the conditions as independent suggests

$$\log n_p \asymp \pi(p) \asymp \frac{p}{\log p},$$

strictly between polynomial and Cambie's $e^{(1+o(1))p}$. The data below (each value re-verified as admissible, and for $p \leq 31$ as minimal by exhaustive search) fit $\log n_p \approx \pi(p)$ well.

| $p$ | $n_p$ | $M$ | $i_0$ | $\log n_p$ | $\log n_p/\pi(p)$ |
|---|---|---|---|---|---|
| 5 | 12 | 2 | 2 | 2.48 | 0.83 |
| 7 | 30 | 4 | 2 | 3.40 | 0.85 |
| 11 | 280 | 25 | 5 | 5.63 | 1.13 |
| 13 | 210 | 16 | 2 | 5.35 | 0.89 |
| 17 | 1456 | 85 | 11 | 7.28 | 1.04 |
| 19 | 612 | 32 | 4 | 6.42 | 0.80 |
| 23 | 2640 | 114 | 18 | 7.88 | 0.88 |
| 29 | 34776 | 1199 | 5 | 10.46 | 1.05 |
| 31 | 8100 | 261 | 9 | 9.00 | 0.82 |
| 37 | 665280 | 17980 | 20 | 13.41 | 1.12 |
| 41 | 1154440 | 28157 | 3 | 13.96 | 1.07 |

**Conjecture.** $\log n_p \asymp p/\log p$ as $p \to \infty$ through primes. In particular $n_p \neq p^{O(1)}$.